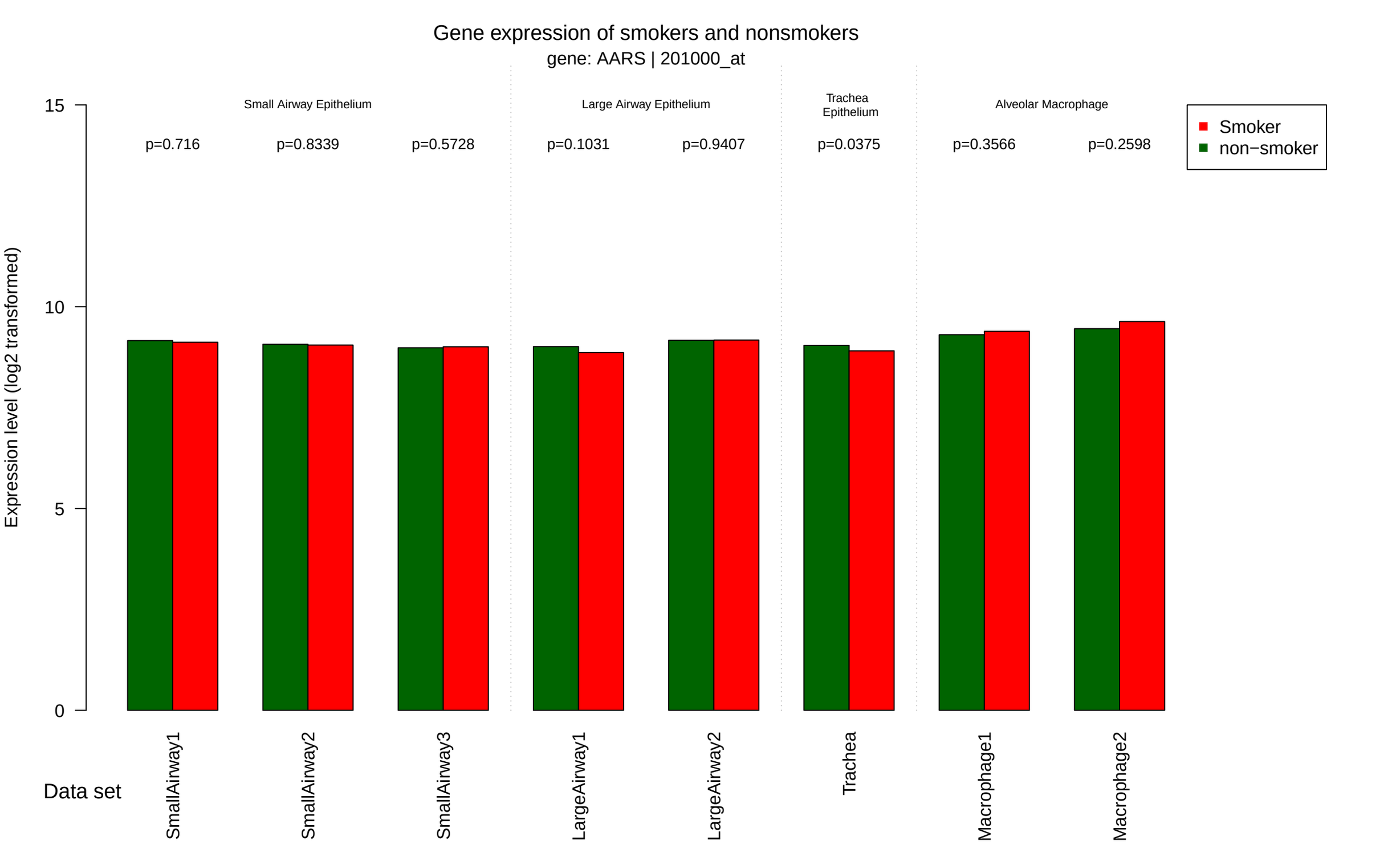
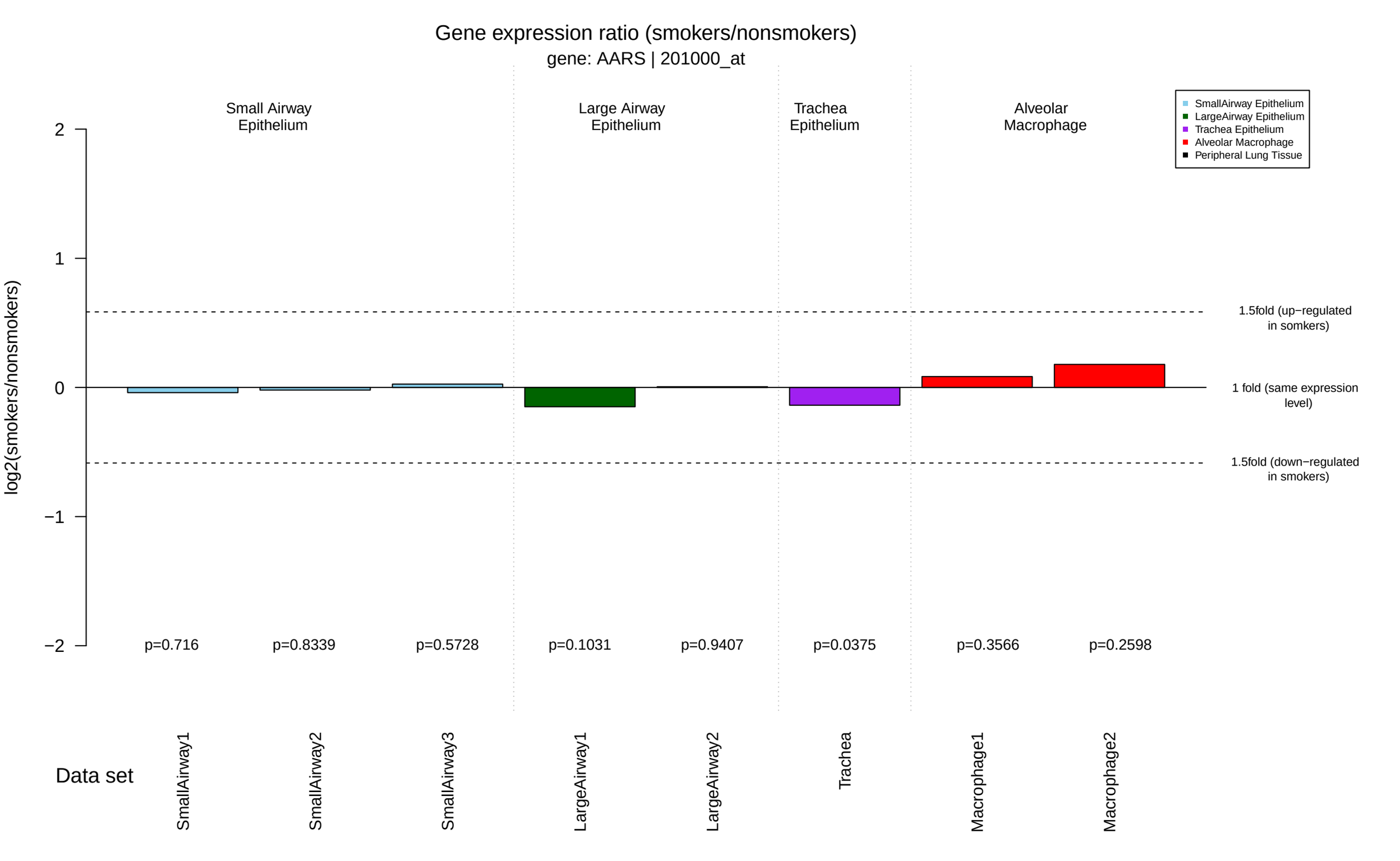
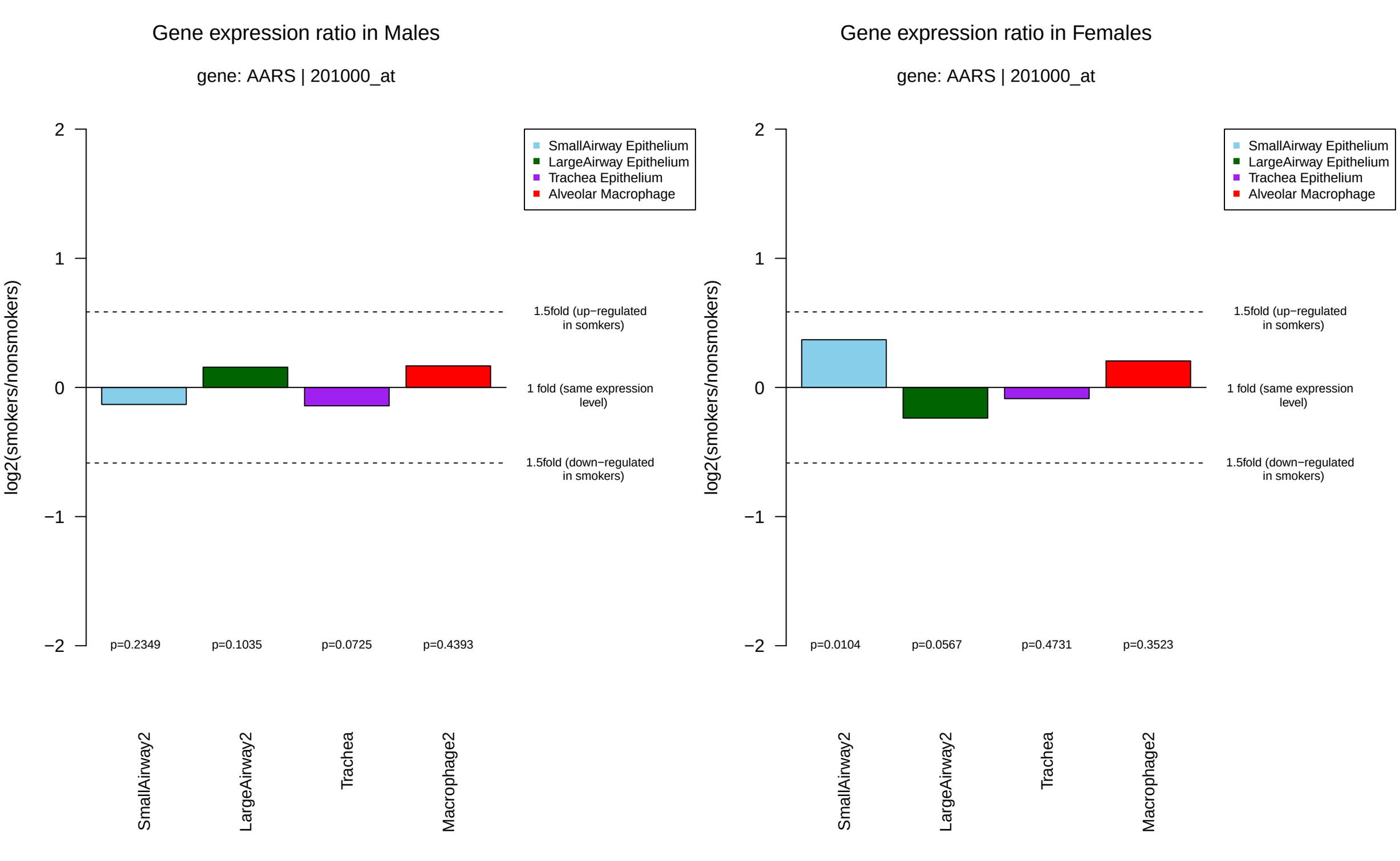
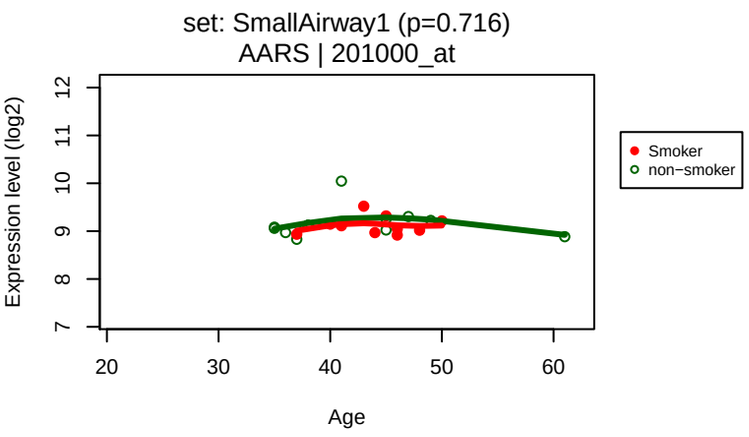
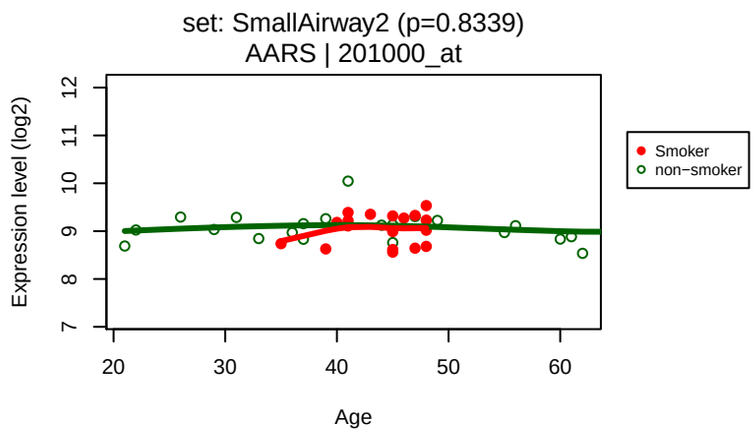
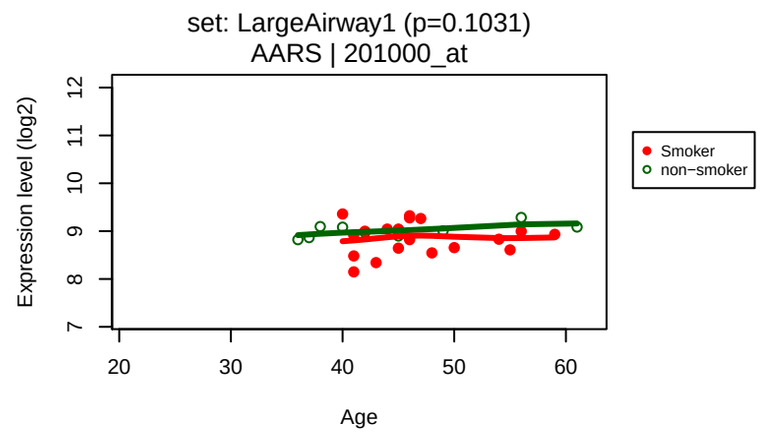

Gene expression of smokers and nonsmokers
gene: AARS | 201000_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.716
p=0.8339
p=0.5728
p=0.1031
p=0.9407
p=0.0375
p=0.3566
p=0.2598
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

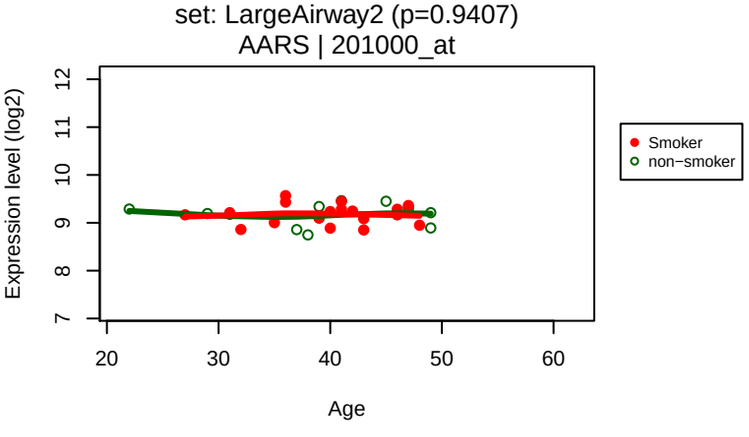

Gene expression ratio (smokers/nonsmokers)
gene: AARS | 201000_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.716
p=0.8339
p=0.5728
p=0.1031
p=0.9407
p=0.0375
p=0.3566
p=0.2598
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

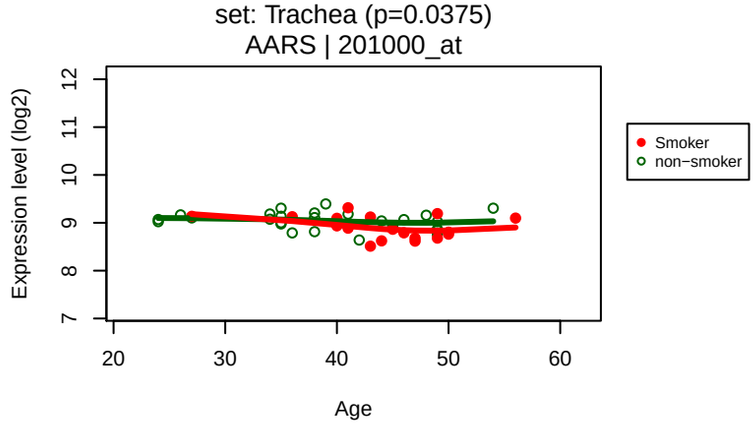

Gene expression ratio in Males
gene: AARS | 201000_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2349
p=0.1035
p=0.0725
p=0.4393
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: AARS | 201000_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0104
p=0.0567
p=0.4731
p=0.3523
SmallAirway2
LargeAirway2
Trachea
Macrophage2

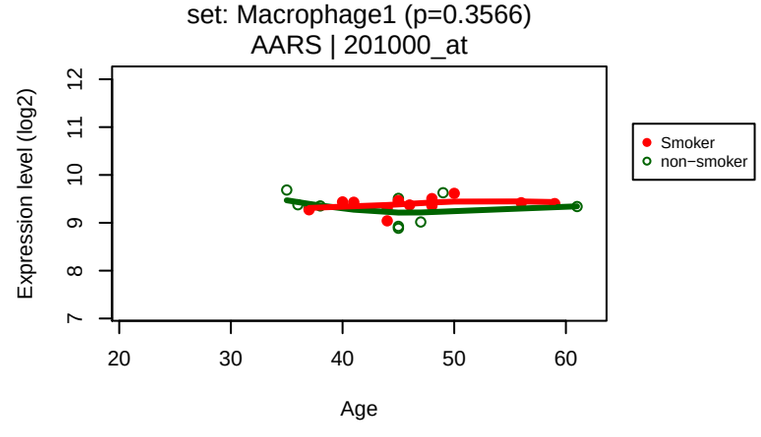

set: SmallAirway1 (p=0.716)
AARS | 201000_at
Expression level (log2)
Age
Smoker
non−smoker

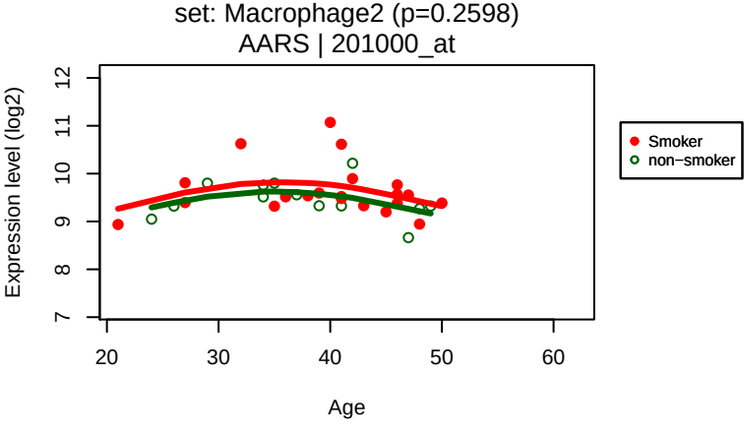

set: SmallAirway2 (p=0.8339)
AARS | 201000_at
Expression level (log2)
Age
Smoker
non−smoker

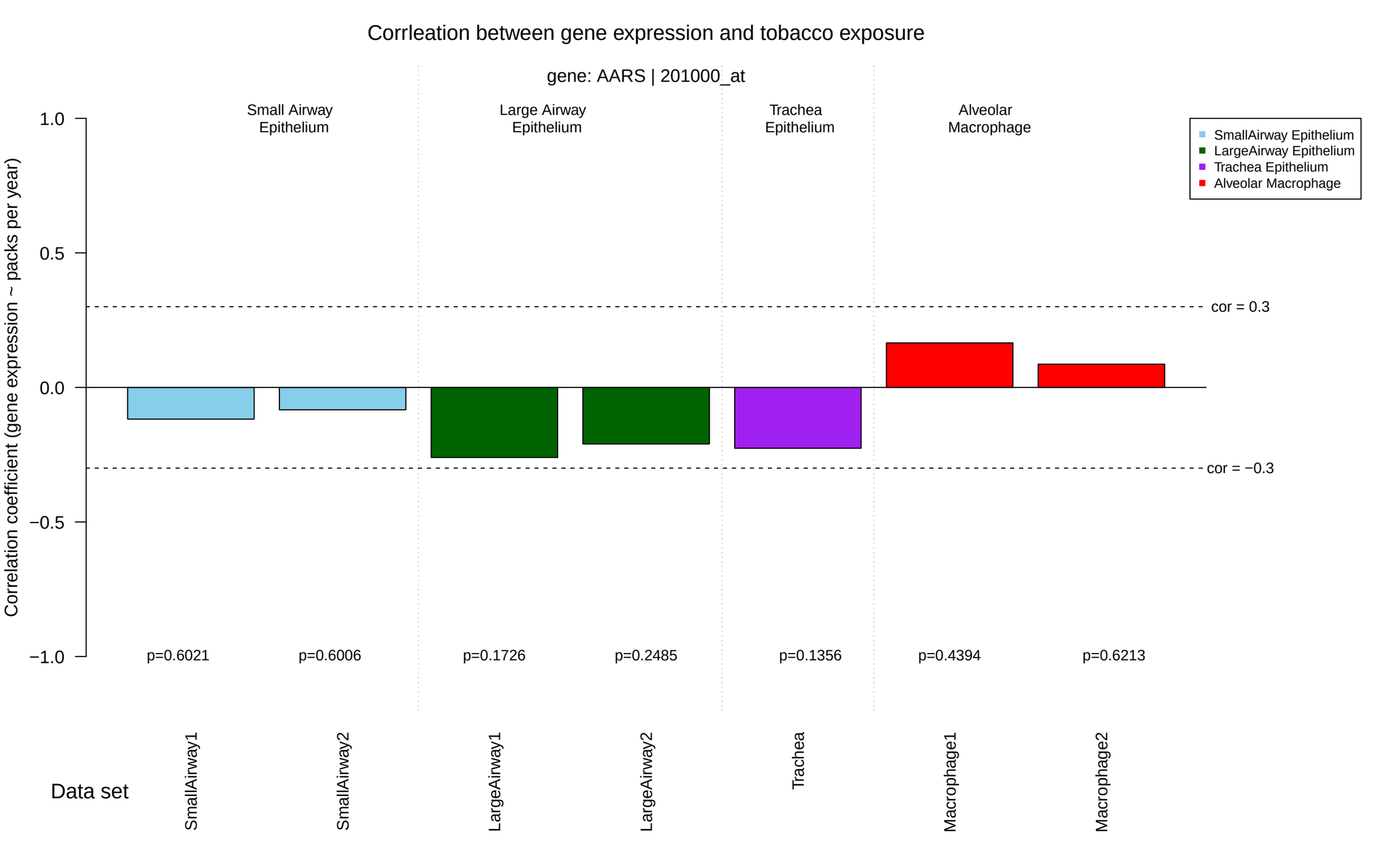

set: LargeAirway1 (p=0.1031)
AARS | 201000_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.9407)
AARS | 201000_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0375)
AARS | 201000_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3566)
AARS | 201000_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2598)
AARS | 201000_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: AARS | 201000_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.6021
p=0.6006
p=0.1726
p=0.2485
p=0.1356
p=0.4394
p=0.6213
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2